CS211 Test 2 Version 2

Total Mark: 30 marks

1) A segmented address space uses paged virtual memory. Each virtual address has a 3 bit segment number, a 2 bit page number and an 13 bit offset within the page. In the following page tables, a "-" indicates that the page is not in memory. There are a total of 16 page frames.

segment 0		segment 1		segment 2		segment 3	
page	frame	page	frame	page	frame	page	frame
0	8	0	-	0	-	0	-
1	1	1	5	1	3	1	4
2	-	2	6	2	2	2	-
3	0	3	-	3	-	3	7

a) Calculate the total number of bits for a logical address. (1 mark)
ANS: **18 bits**

b) Calculate the total number of bits for a physical address. (1 mark)
ANS: **17 bits.**

c) What is the highest hexadecimal virtual address in segment 1 that will cause a page fault? (2 marks)
001 11 1 1111 1111 1111 = $\boldsymbol{FFFF_{16}}$

d) Find the real memory address (in hexadecimal) if the virtual hexadecimal address is BC12. (2 marks)
BC12 = 101 1 1100 0001 0010 → Segment 1 Page 1, -> frame 5
Physical address = 101 1 1100 0001 0010 = $\boldsymbol{BC12_{16}}$

2) A register has contents in hexadecimal of 33ABCDEF. After a right **arithmetic** shift of 12 bits, what are the contents of the register? Give your answer in hexadecimal. (2 marks)
Answer:
0011 0011 1010 1011 1100 <u>1101 1110 1111</u> ½ mark

0000 0000 0000 0011 0011 1010 1011 1100 1 mark

=> **00033ABC** ½ mark {important to list all 8 hexadecimal values here including zero's}

3) Justify whether it is possible to design an expanding opcode to allow the following to be encoded in a 20-bit instruction? An address is 5 bits. (5 marks)
15 instructions with three addresses
14 instructions with two addresses
31 instructions with one address
16 instructions with 0 address

Answer:
00000 R1 R2 R3
…. 1 mark
01110 R1 R2 R3

01111 00000 r2 r3

… 1 mark

01111 01101 r2 r3

01111 01110 00000 r3

…. 1 mark

01111 01111 01110 r3

01111 01111 01111 00000

…… 1 mark

01111 01111 01111 01111

Therefore, this design is possible. 1 mark

{a different justification technique was accepted this time with partial marks but in future, answer this question by designing the expanding opcode}

4) Suppose a CPU has 4 data pins. How many operations will it take to read a 32-bit word. (2 mark)
ANS: 8 operations.

5) Explain whether pipelining reduces the execution time of a program. (2 marks)

Yes: Each instruction will still take several cycles to pass through the pipeline completely and in parallel.

6) A processor uses a fixed-length 36-bit instruction format for a one address machine. The processor has 129 instructions.
   a) Calculate the number of bits for opcode. (1 mark)
   b) Calculate the number of bits for operand. (1 mark)
   c) Calculate the maximum memory address that a program can in this machine can access. (2 marks)
   d) Is it possible to expand the instruction set of this machine? Explain your answer. (2 marks)

Ans:

a) 8 bits
b) 28 bits
c) 268435456 or 2^28
d) Yes. There is space for 127 instructions.

7) Given the following equation:

$$Y = (A - (B / C)) X (D + E)$$

   a) Write a program to solve the equation using a one-address machine with LOAD, STORE, MUL, ADD, DIV, and SUB instructions. The machine has variable-length instructions. LOAD and STORE instructions have 8 bits of opcode while other instructions have 6 bits of opcode. Operands are of fixed 16 bits. Y,A,B,C,D and E are labels representing the content

of memory locations. A,B,C,D and E must NOT be overwritten. You may use other memory locations (eg. T) as temporary storage. (5 marks)

b) Calculate the length of your program in bits. (2 marks)

ANS: a)

LOAD B

DIV C

STORE T

LOAD A

SUB T

STORE T

LOAD D

ADD E

MUL T

STORE Y

B) 6 X 24 + 4 X 22 = 232 bits